

# 20251006 rMZR in HII But No NGC4383

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Note.

1. Try removing the abnormal NGC4383. The reason is that it almost occupies the entire independent parameter space in rMZR. So if we want to study the SFR dependence in rMZR, it will be biased by this single galaxy. Therefore, we can try removing it and see how the results change.
2. Only spaxels from HII regions are included.
3. Update to 14  $\Sigma_{\text{SFR}}$  bins, with binwidth 0.25 dex
4. Stick with these 4 indicators: D16, PG16, O3N2-M13 and O3N2-C20.

Summary:

**Let's focus on O3N2 first (third and fourth rows).**

Kind of different SFR dependence in rMZR, maybe due to that indicators are more dependent on O abundance or N abundance?

1. For D16 and PG16: higher  $\Sigma_{\text{SFR}}$  at higher metallicity up to at least  $\Sigma_* \lesssim 8.5$ .
2. For both O3N2, we can see there is an inversion point at  $\Sigma_* \sim 7.8$  (lower metallicity shows higher  $\Sigma_{\text{SFR}}$  in low mass end; while metallicity positive-correlates with  $\Sigma_{\text{SFR}}$  in high mass end).

For individual galaxy, based on their positions in  $Z$ - $\Sigma_*$  space of HII spaxels, they can be separated into the following 4 classes: "low" (primarily less than the vertical line at  $\Sigma_* = 7.8 M_\odot/\text{kpc}^2$ ), "both" (sitting across the line) and "high" (primary high than the line), or "few" (no enough spaxels to tell the position).

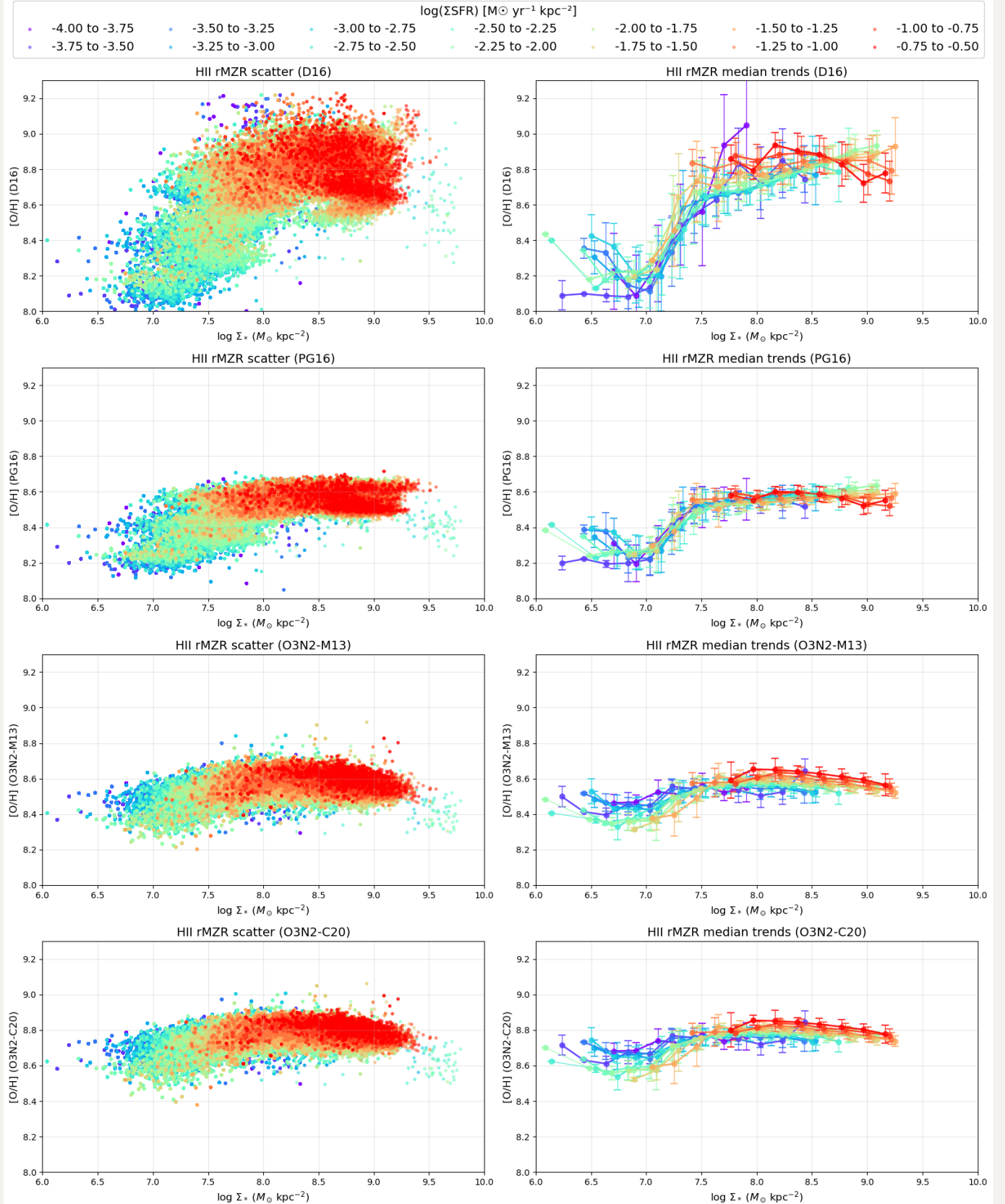
And in other way, based on the correlation between  $Z$  and  $\Sigma_{\text{SFR}}$  at fixed  $\Sigma_*$ , they can be also separated into 3 classes if they are not "few": "positive", "negative" or "inverse" ("negative" in low mass end while "positive" in high mass end, with inversion point at  $\Sigma_* \sim 7.8 M_\odot/\text{kpc}^2$ ).

Thus, we have the following table (\* are those "abnormalies" in rMZR, together with the removed NGC4383):

| ID       | POSITION     | CORRELATION        | EXPECTATION |
|----------|--------------|--------------------|-------------|
| IC3392   | High         | Positive           | Yes         |
| NGC4064  | High         | Positive           | Yes         |
| NGC4192  | Cross        | Inverse            | Yes         |
| NGC4293  | Few (High?)  | -                  | -           |
| NGC4298  | Cross        | Positive (mostly?) | No?         |
| NGC4330* | Low          | Negative           | Yes         |
| NGC4396* | Low          | Negative           | Yes         |
| NGC4419  | High         | Positive           | Yes         |
| NGC4457  | Few (High?)  | -                  | -           |
| NGC4501  | Cross        | Positive           | No          |
| NGC4522* | Low          | - (Positive?)      | No?         |
| NGC4694* | High         | Positive           | Yes         |
| NGC4698  | Few (Cross?) | -                  | -           |

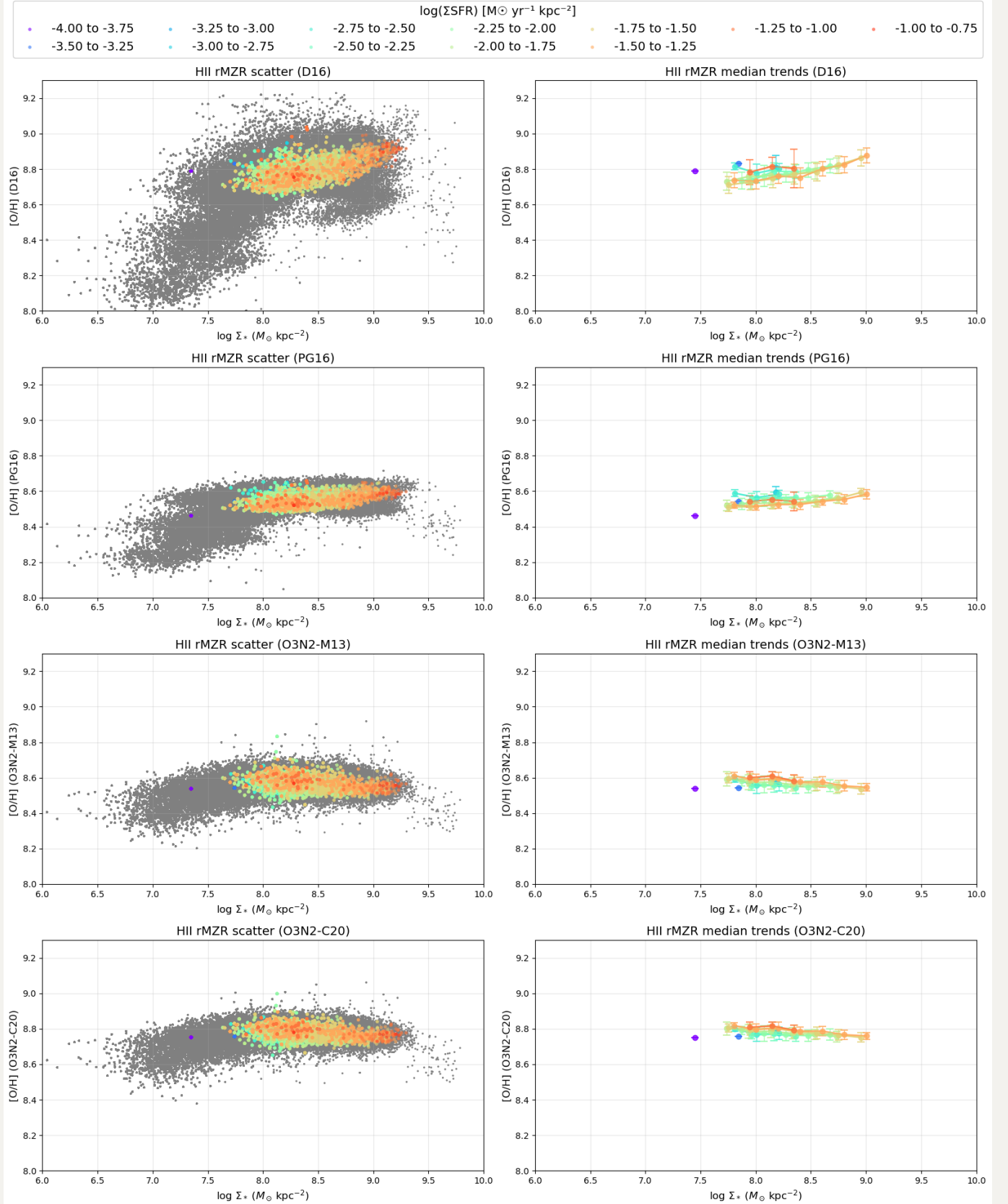
## Combined

rMZR behavior in  $\Sigma_{\text{SFR}}$  bins: HII regions only, No {'NGC4383'} (Min 50 datapoints, Adjacent bins only)

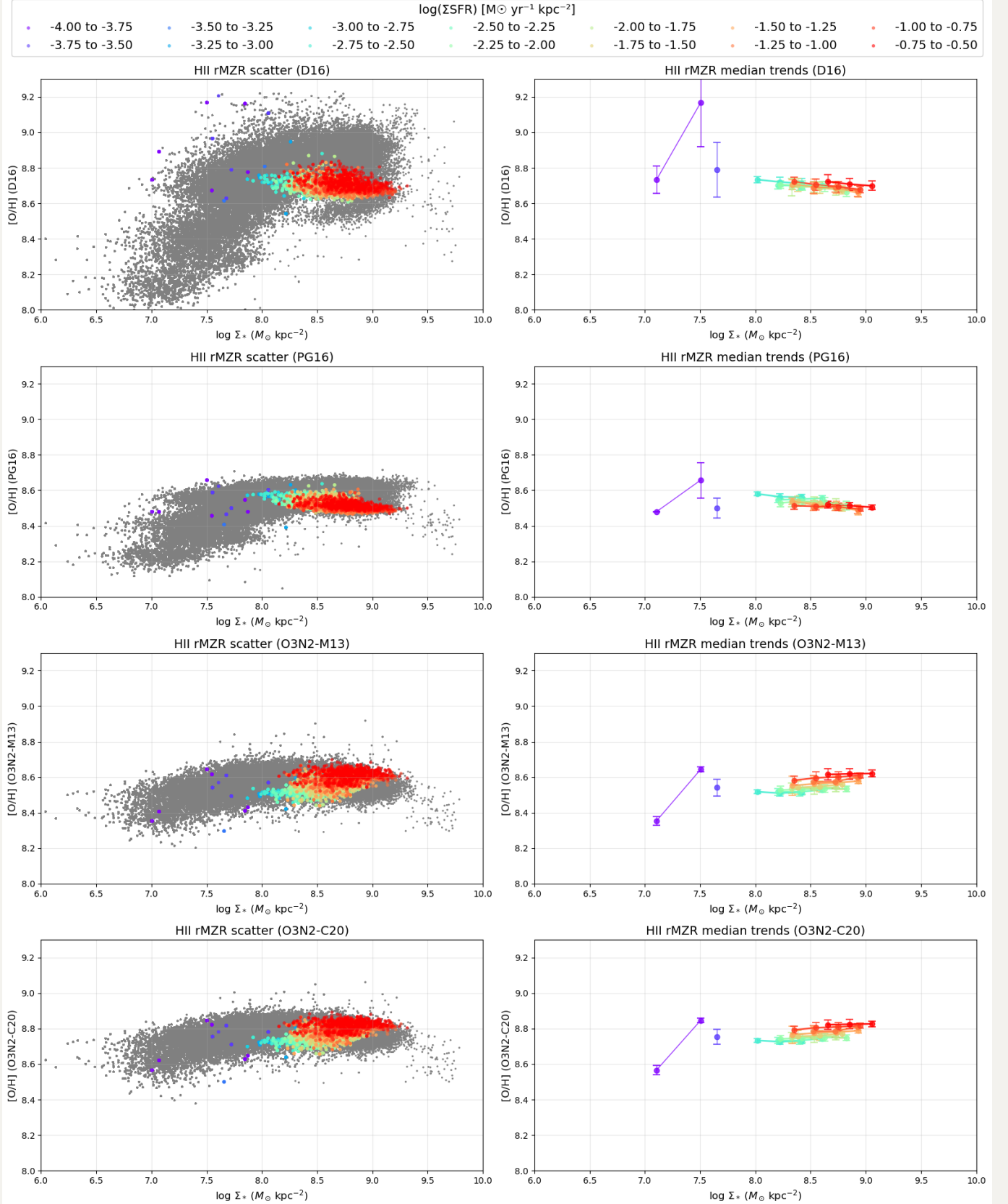


Individual

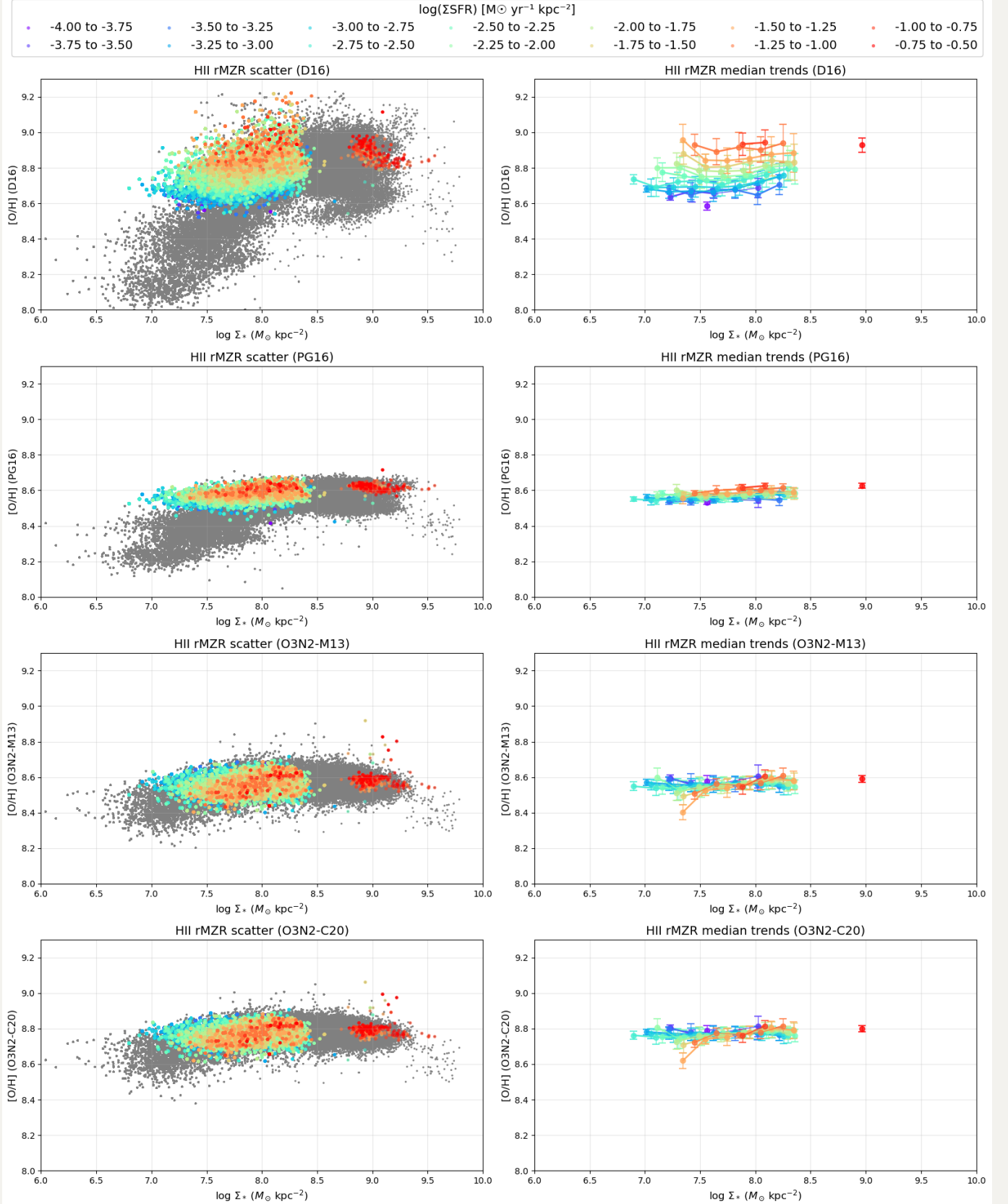
IC3392: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



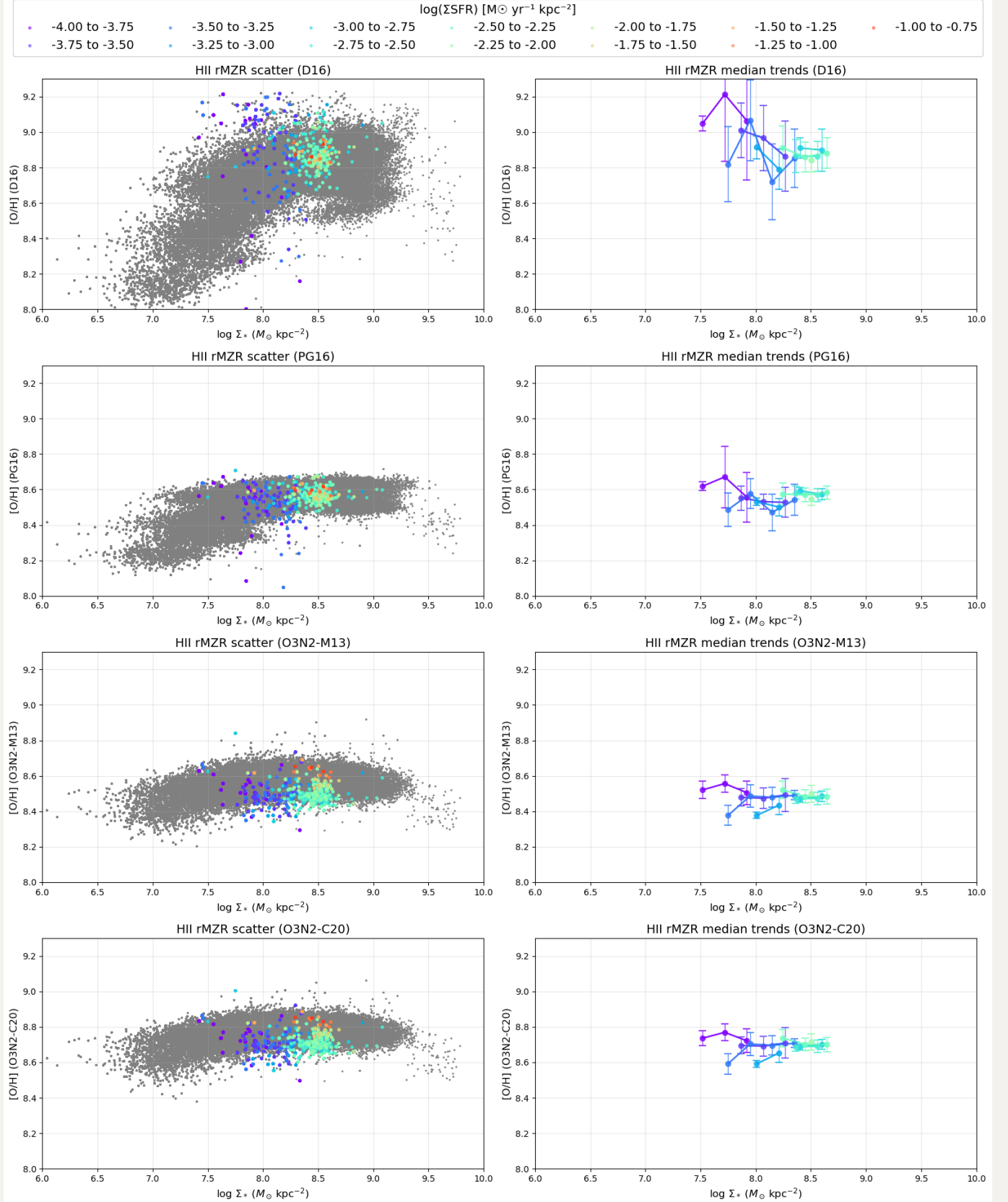
NGC4064: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



NGC4192: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)

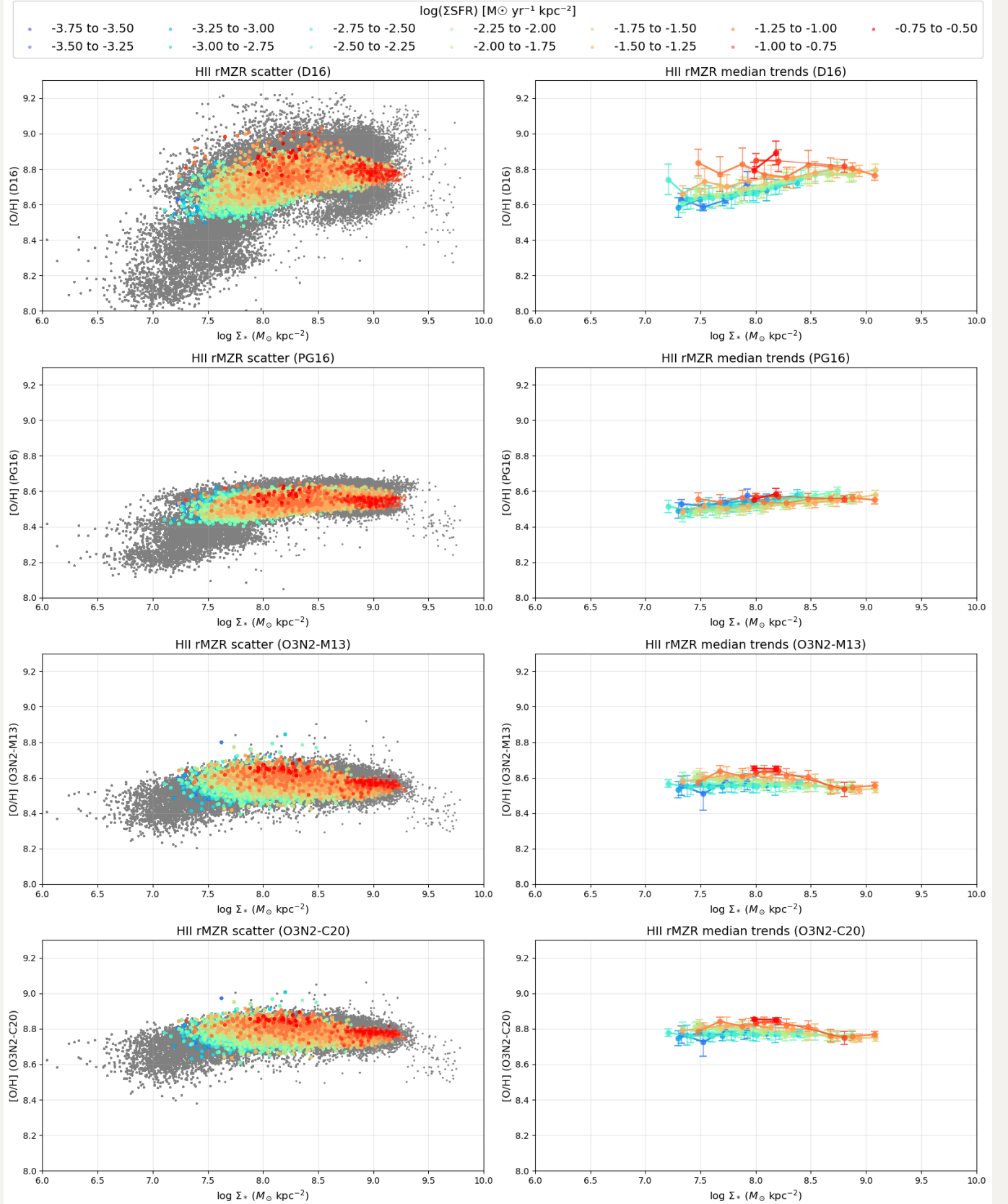


NGC4293: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



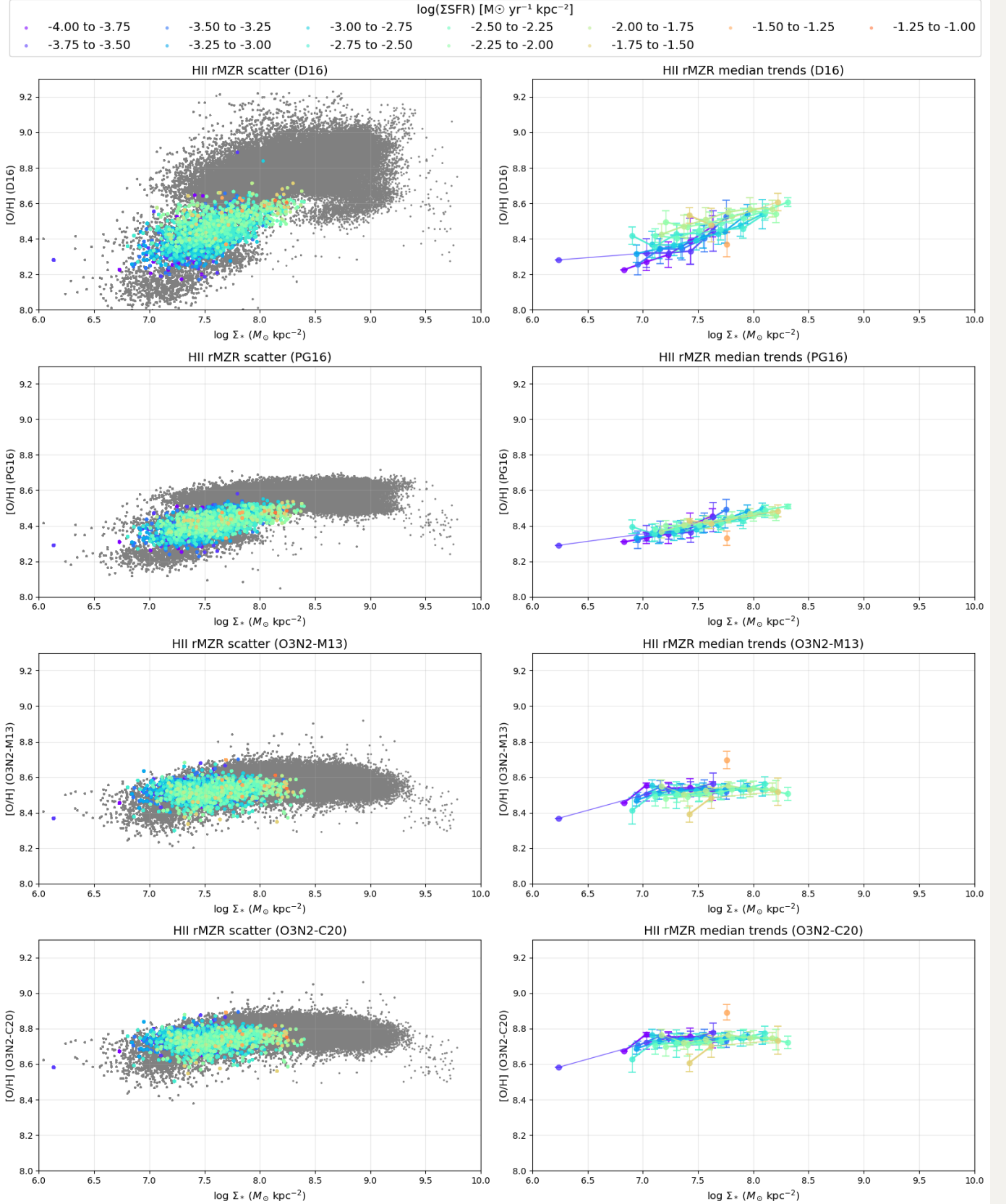


NGC4298: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)

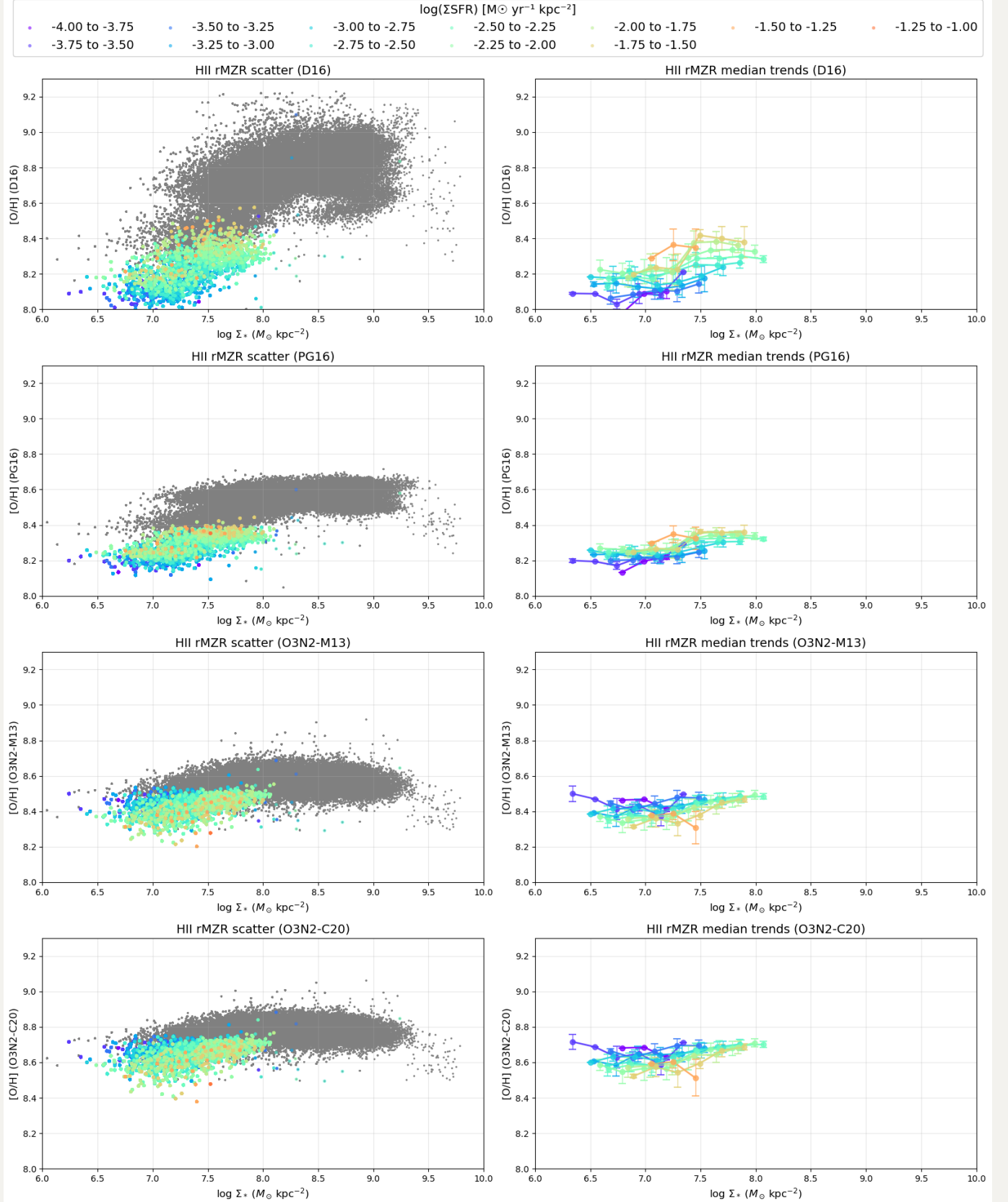




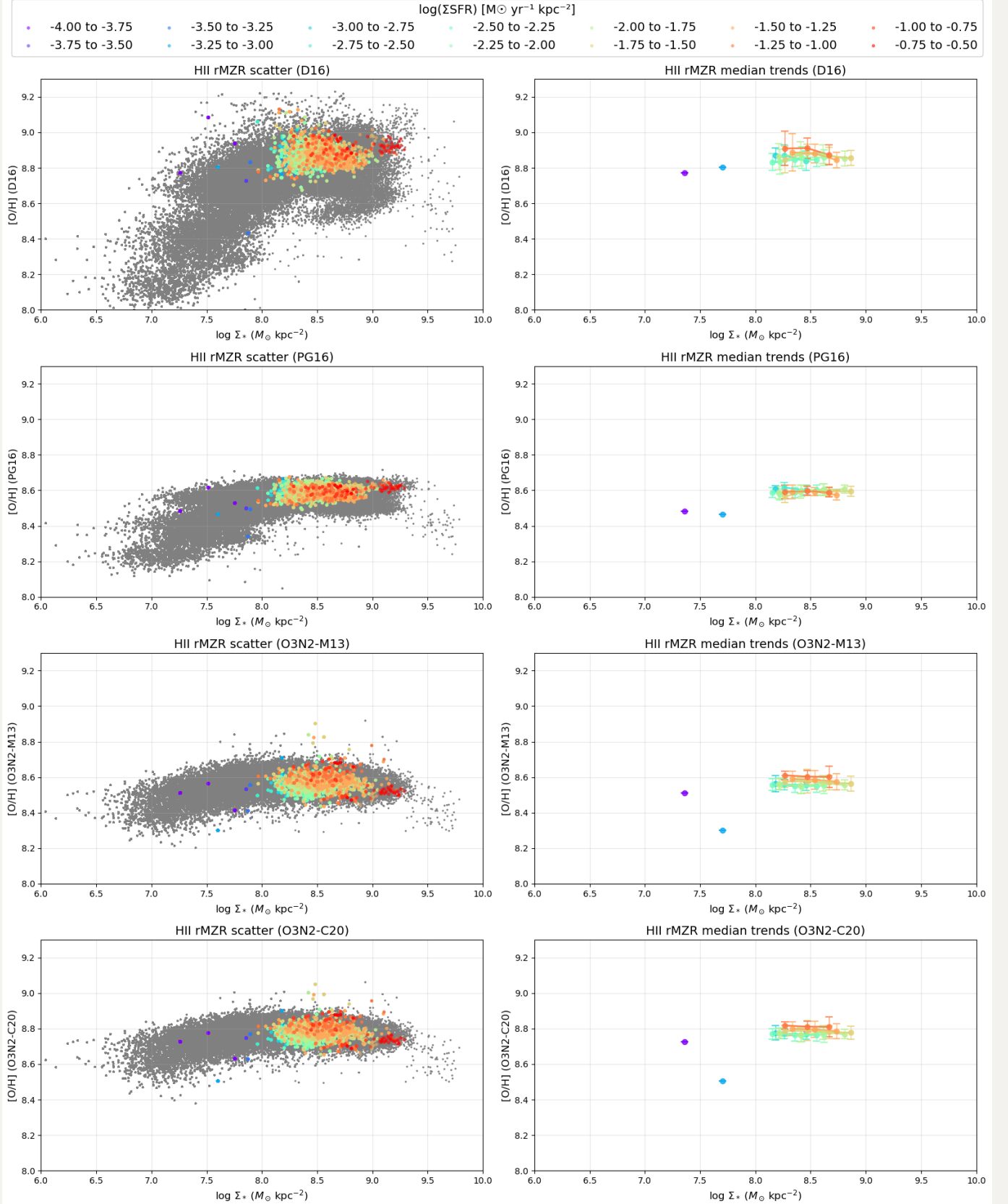
NGC4330: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



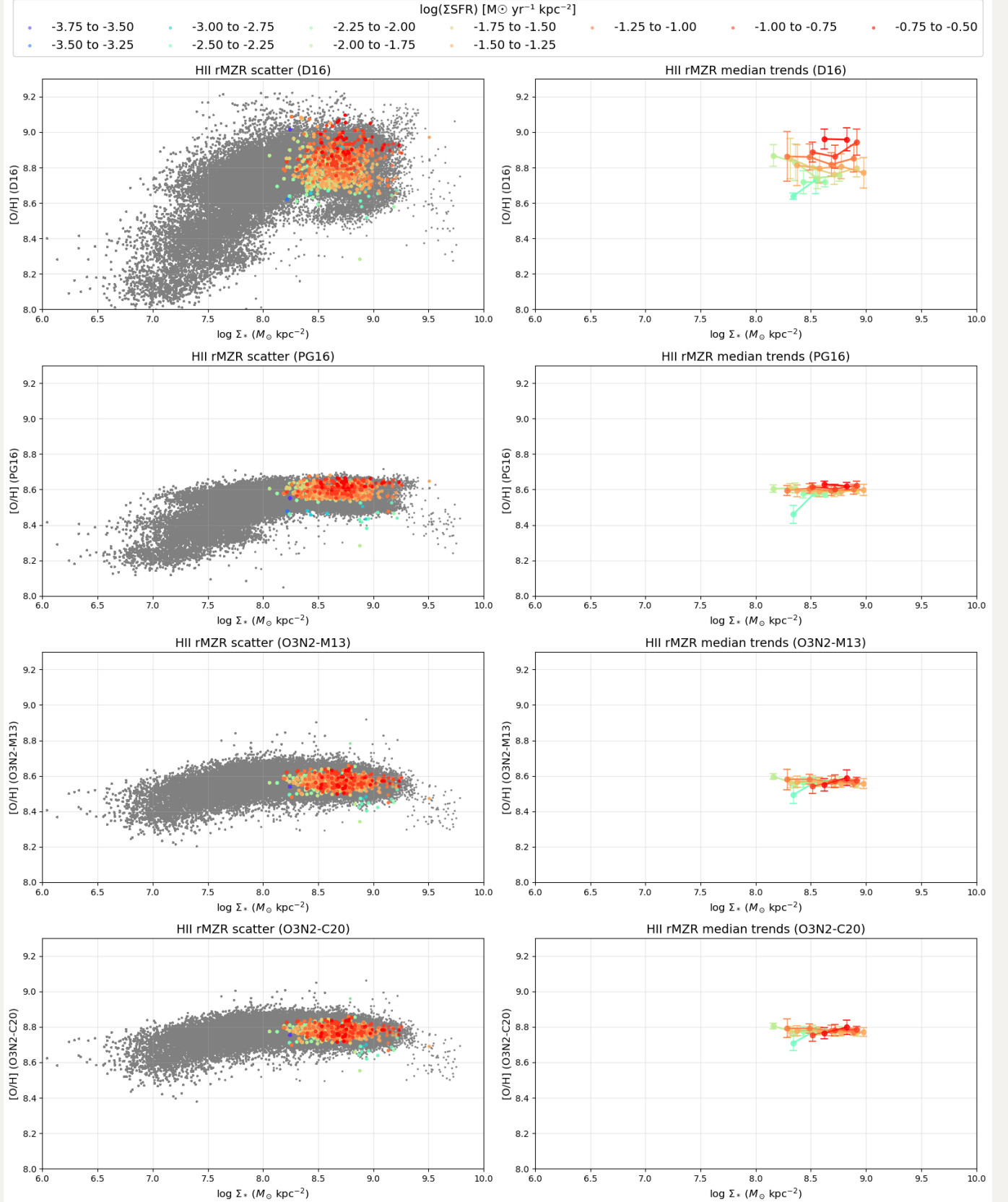
NGC4396: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



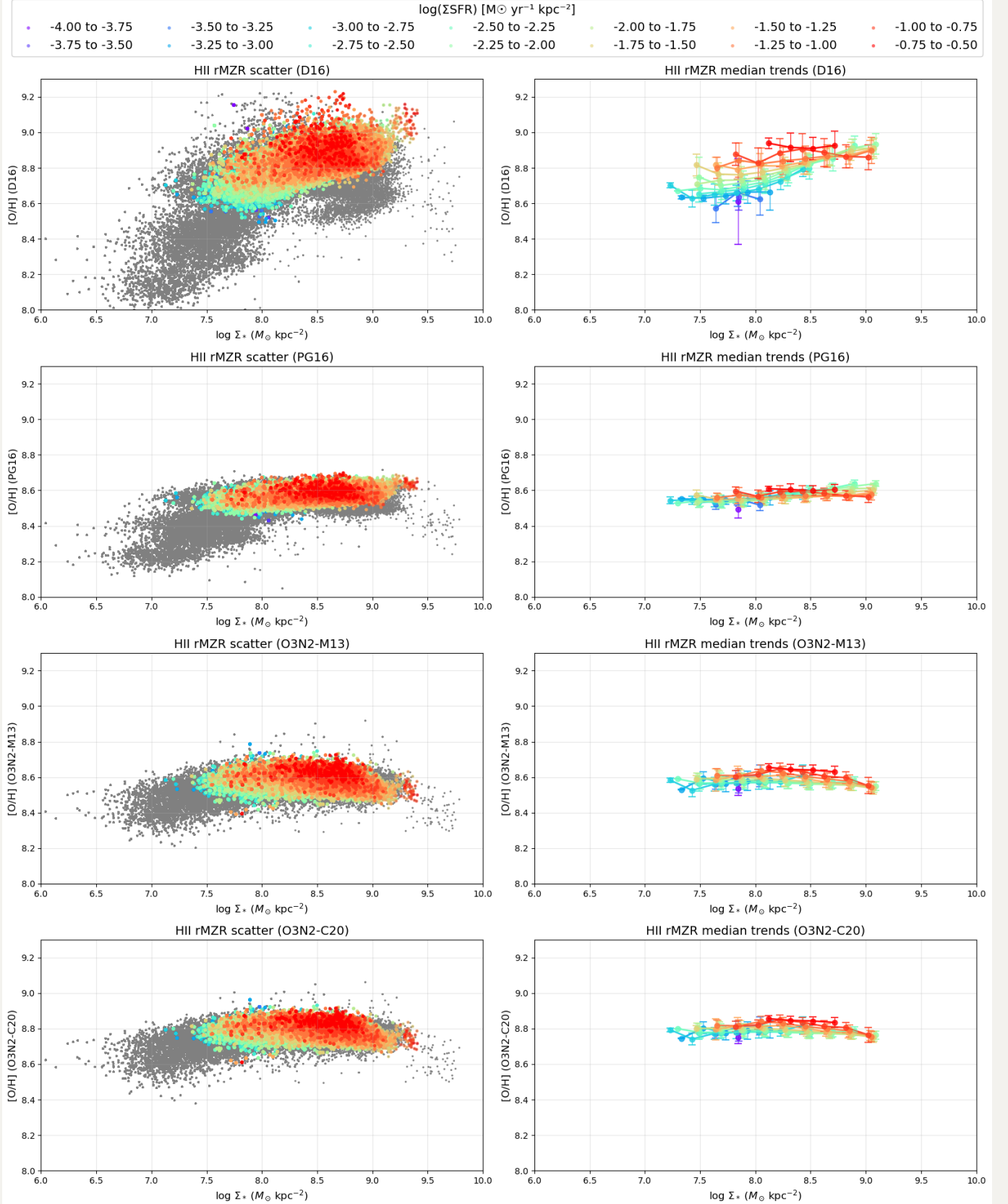
NGC4419: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



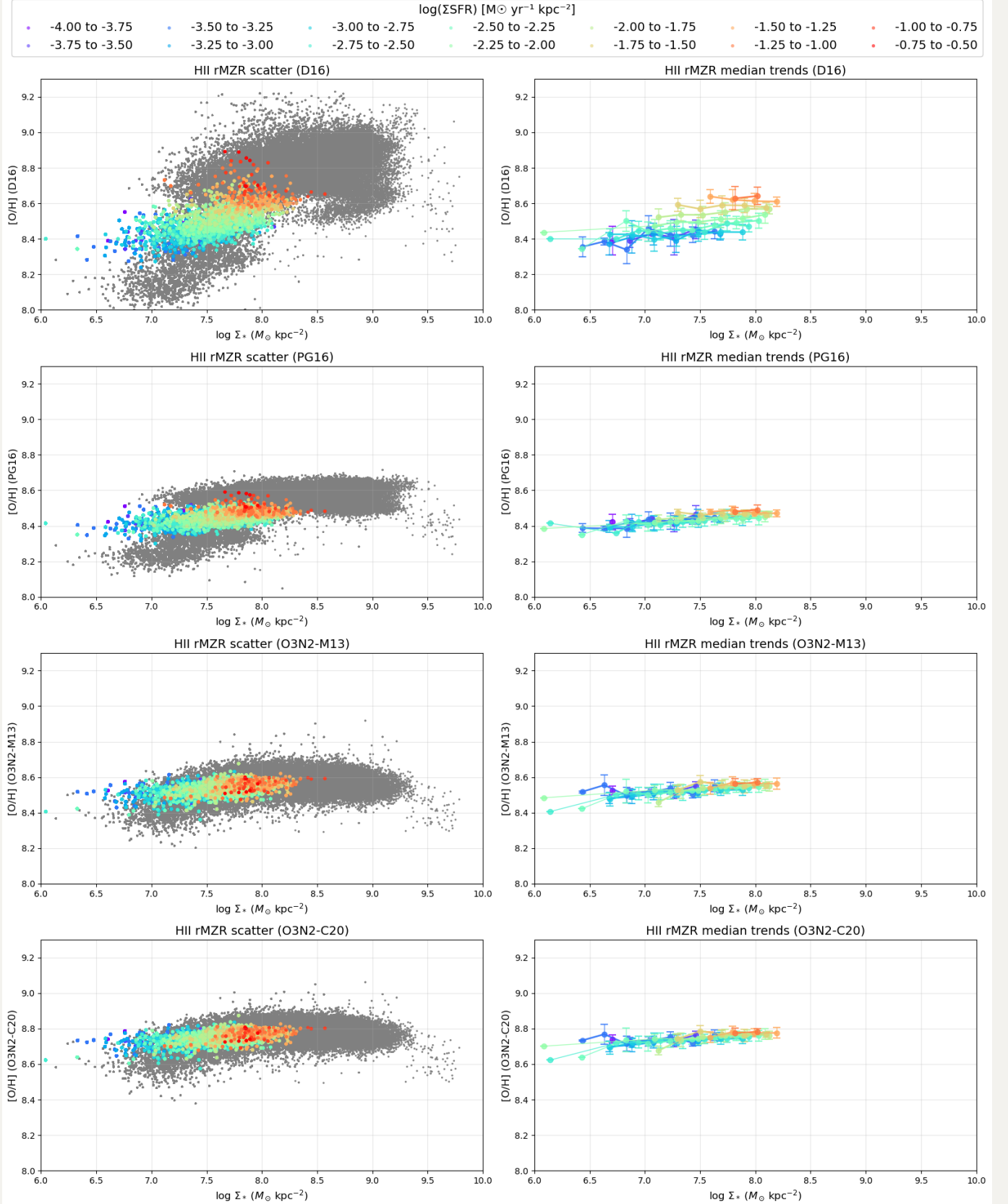
NGC4457: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



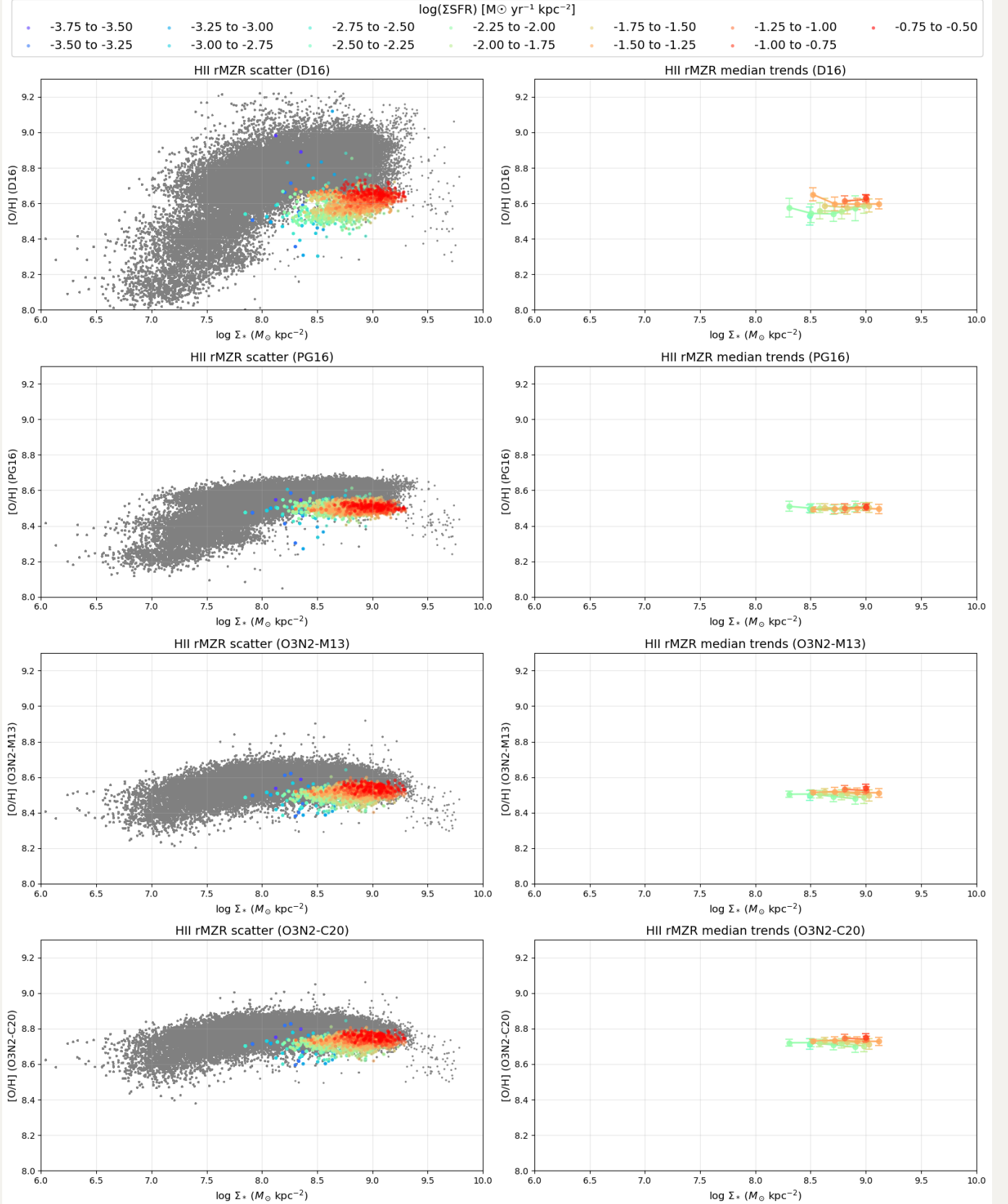
NGC4501: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



NGC4522: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)



NGC4694: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)





NGC4698: rMZR behavior in  $\Sigma_{\text{SFR}}$  bins - HII regions only (Min 50 datapoints, Adjacent bins only)

